

Multi-Agent Systems for Sustainable Smart Farming

A Systematic Literature Review of Cooperative Irrigation, Resource Allocation, Livestock Management, and Environmental Sustainability

1. Background and Motivation

Modern agriculture requires simultaneous decisions concerning irrigation, crop management, livestock welfare, weather variability, energy use, and the allocation of limited resources. These processes are strongly interconnected. Multiple crop zones may require irrigation while livestock simultaneously require water; rainfall forecasts may change irrigation priorities; drought and heatwaves may increase competition for shared resources; and farm-management decisions can influence water consumption, energy use, greenhouse-gas (GHG) emissions, soil quality, and other environmental outcomes.

These characteristics make agriculture a suitable domain for **Multi-Agent Systems (MAS)**, in which autonomous and heterogeneous agents perceive local conditions, pursue individual or shared objectives, communicate, cooperate, and negotiate over resources. Existing research has investigated multi-agent irrigation systems, water-allocation negotiation, intelligent agricultural agents, learning-based irrigation control, and precision livestock systems. However, these areas are often studied separately rather than as components of an integrated farm-level decision problem.

The sustainability dimension is particularly important. Agriculture accounts for a major share of global freshwater withdrawals, making irrigation efficiency and water scarcity central concerns. Agricultural activities also contribute substantially to greenhouse-gas emissions through sources such as livestock methane, manure management, fertiliser use, energy consumption, and land-use practices. Other environmental concerns include nutrient runoff, water pollution, soil degradation, and pressure on ecosystems.

This project therefore proposes a **Systematic Literature Review (SLR)** investigating how Multi-Agent Systems have been applied to integrated smart-farm decision-making, with particular emphasis on **irrigation, shared-resource allocation and negotiation, livestock management, and environmental sustainability**.

2. Aim and Research Contribution

The main objective is to systematically identify, classify, and critically analyse existing research on MAS-based agricultural decision systems and determine:

How are Multi-Agent Systems designed and used to coordinate agricultural decision-making and shared-resource allocation across crop, irrigation, and livestock systems, and to what extent do existing approaches address environmental sustainability?

The review will analyse the literature through the following connected dimensions:

Agent architecture → communication and coordination → resource management → agricultural outcomes → environmental outcomes

The intended contribution is a **taxonomy and evidence map of agricultural MAS**, identification of technical and research gaps, and an evidence-based conceptual architecture for a future integrated sustainable smart-farm MAS that may subsequently form the foundation for thesis work.

3. Research Questions

Primary Research Question

RQ0. How have Multi-Agent Systems been designed, implemented, and evaluated for coordinated decision-making in smart agriculture, particularly for irrigation, shared-resource allocation, livestock management, and environmental sustainability?

Secondary Research Questions

RQ1 — Applications and Agent Roles:

What agricultural problems are addressed by MAS, and which agent roles are used, such as farm-manager, weather, water/resource, crop/field, livestock, pasture, sensor, or machinery agents?

RQ2 — Agent Architecture and Intelligence:

Which agent architectures and decision mechanisms are used, including reactive or rule-based agents, utility-based agents, BDI architectures, optimisation, planning, machine learning, reinforcement learning/MARL, model-predictive control, and emerging LLM-based agents?

RQ3 — Communication and Coordination:

How do agricultural agents communicate, cooperate, and coordinate, and which mechanisms are used for message exchange, task allocation, conflict resolution, and distributed decision-making?

RQ4 — Resource Allocation and Negotiation:

How do MAS manage scarce or competing resources such as water, feed, pasture, energy, land, or machinery? Which negotiation, auction, priority, utility, contract-net, or optimisation mechanisms are employed?

RQ5 — Irrigation and Water Scarcity:

How are soil moisture, crop requirements, rainfall forecasts, drought, and reservoir availability incorporated into irrigation decisions, and what evidence exists regarding water savings, allocation efficiency, and resilience to scarcity?

RQ6 — Livestock Management and Welfare:

How are livestock represented within intelligent or multi-agent agricultural systems, including water and feed demand, heat stress, health, grazing or pasture allocation, and animal welfare?

RQ7 — Environmental Sustainability:

Which environmental impacts are explicitly modelled or evaluated, including:

- freshwater use and water scarcity;
- GHG emissions such as CO₂, CH₄, and N₂O;
- livestock methane and manure-related emissions;
- irrigation and farm energy consumption;
- fertiliser and nutrient runoff;
- soil degradation and soil health;
- land and pasture use;
- pollution and ecosystem impacts?

RQ8 — Environmental Mitigation and Trade-offs:

How are agents used to reduce environmental impacts through approaches such as precision irrigation, demand-aware water allocation, efficient feeding, manure or fertiliser management, renewable-energy scheduling, pasture management, or carbon-aware decision-making, and how are trade-offs between productivity, animal welfare, and sustainability handled?

RQ9 — Evaluation and Validation:

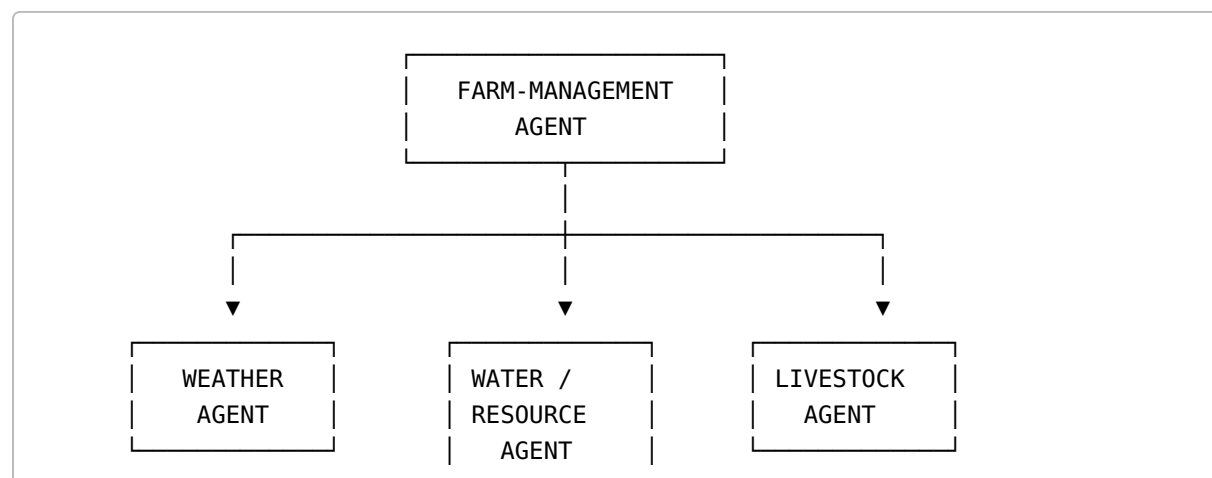
Which datasets, simulations, IoT or sensor systems, field experiments, scenarios, baselines, and metrics are used to evaluate agricultural MAS, and how often are systems validated in realistic or real-world environments?

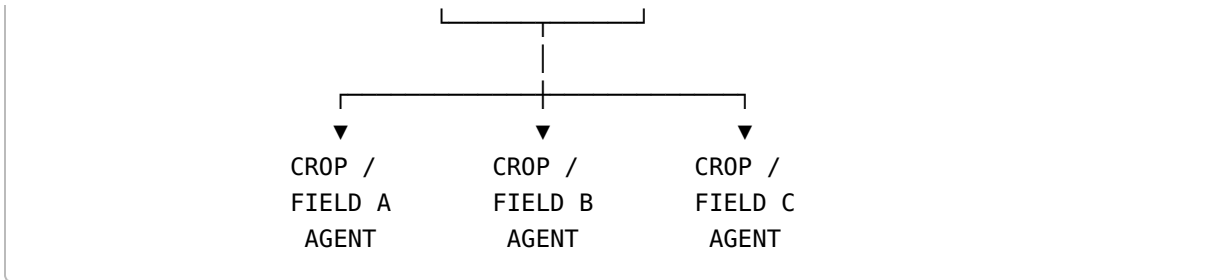
RQ10 — Research Gaps and Future Systems:

Which limitations remain regarding integration, scalability, uncertainty, explainability, interoperability, environmental modelling, and real-world deployment, and what requirements emerge for a future integrated **Multi-Agent Smart Farm Decision System**?

4. Conceptual Scope of the Review

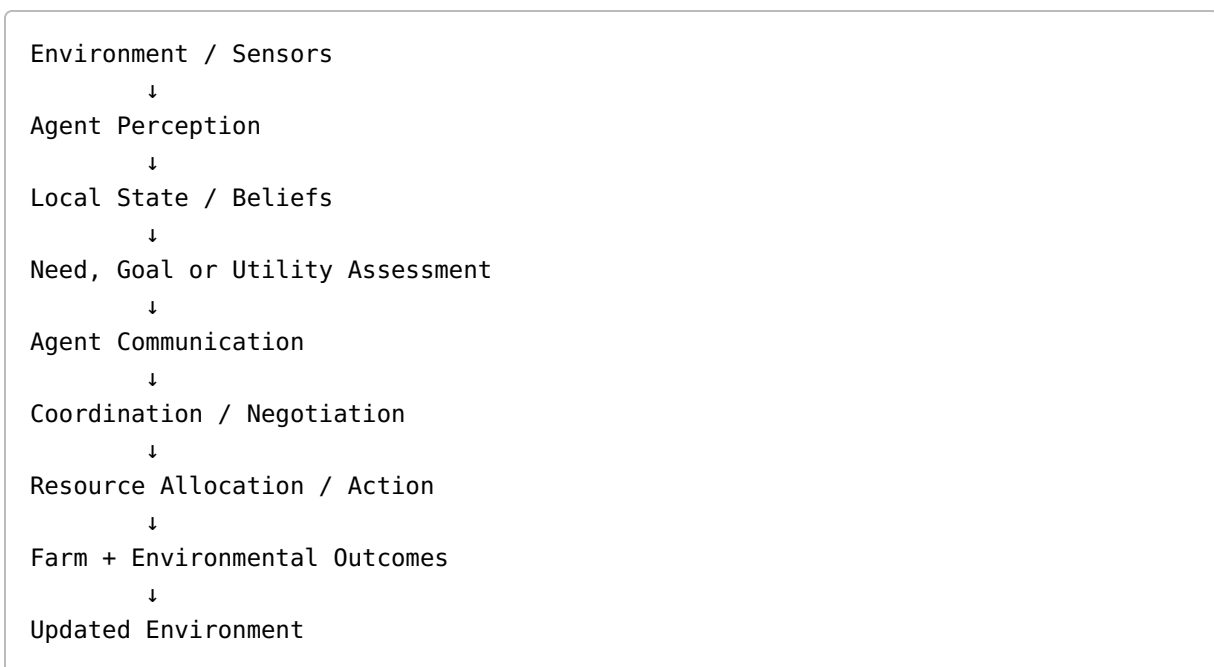
The original smart-farm architecture will be retained as an analytical framework for classifying the literature:





The review will determine which components of this integrated architecture already appear in the literature, which are normally studied independently, and which interactions remain insufficiently explored.

The general MAS decision cycle will also be used as a classification framework:



Particular attention will be given to structured interactions such as resource requests, offers, acceptance or rejection, alerts, information exchange, cooperation, and negotiation.

5. Systematic Review Methodology

The review will follow a predefined and reproducible protocol and will use **PRISMA 2020** to structure and report identification, screening, eligibility, and study inclusion.

Information Sources

The main databases will include:

- Scopus

- Web of Science
- IEEE Xplore
- ACM Digital Library
- ScienceDirect
- SpringerLink

Google Scholar and backward/forward citation searching may be used as supplementary snowballing mechanisms.

Indicative Master Search Strategy

```
("multi-agent system*" OR "multi agent system*" OR
"multi-agent" OR "intelligent agent*" OR
"agent-based decision*")
AND
(agricultur* OR farm* OR "smart farm*" OR
"precision agriculture" OR irrigation OR
livestock OR "precision livestock")
AND
(coordination OR cooperation OR negotiation OR
"resource allocation" OR decision* OR water OR
sustainab* OR environment*)
```

This represents the **conceptual master search strategy** rather than a single literal query to be copied into every database.

Database-Specific Syntax Adaptation

The exact syntax will be adapted to each search engine while preserving the same underlying concepts. Database-specific differences in wildcard support, field names, phrase searching, and Boolean syntax will therefore be documented.

For example, where wildcard restrictions make terms such as:

```
sustainab*
```

inappropriate, they may be expanded explicitly as:

```
(sustainability OR sustainable)
```

Similarly:

agricultur*

may be translated into explicit equivalent terms where required.

The final report will record the **exact query executed in each database, search date, searchable fields, and number of retrieved records**, supporting reproducibility.

Publication Window

The planned publication period is:

2010–October 2026

This window captures contemporary smart-agriculture and MAS research while maintaining a manageable and relevant corpus. Backward citation searching may additionally identify earlier foundational studies when necessary.

Screening Threshold

The initial search will aim to maximise reasonable coverage while maintaining relevance.

If the **deduplicated corpus exceeds approximately 500 records**, the search will be refined by restricting queries specifically to the closest available equivalents of:

Title, Abstract, and Keywords

in each database, while retaining the same conceptual search terms.

This rule will be specified in the review protocol before final screening so that any refinement is systematic rather than an ad-hoc response to the results.

6. Eligibility, Deduplication, and Screening

Studies will generally be included when they:

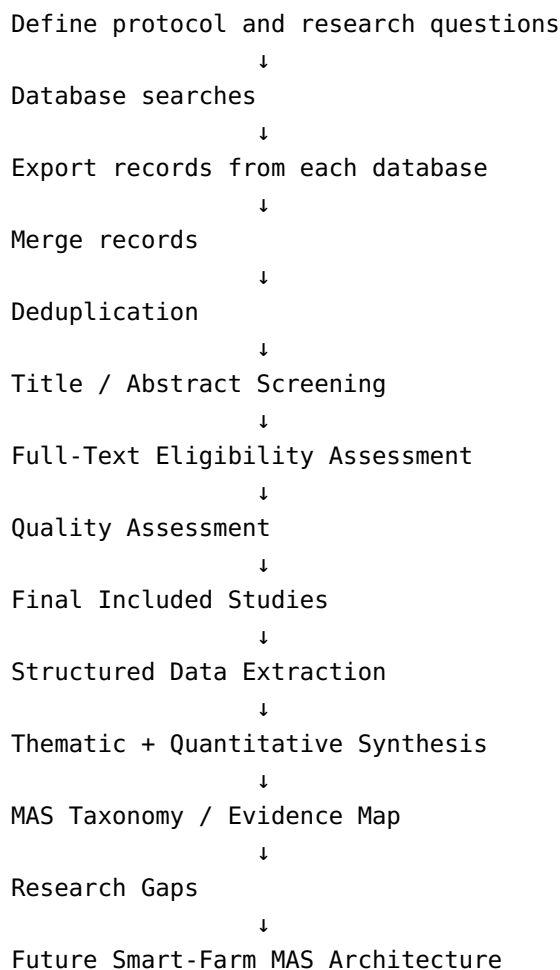
- are peer-reviewed journal articles or conference papers;
- concern agriculture, farming, irrigation, or livestock;
- explicitly use multiple interacting/autonomous agents or a clearly defined MAS or agent-based decision architecture;
- address decision-making, coordination, cooperation, negotiation, or resource management;
- provide sufficient methodological information for analysis;
- are available in English.

Studies will generally be excluded when they:

- only implement IoT monitoring without meaningful agent autonomy;
- use conventional ML prediction without multi-agent interaction;
- use sensors only for data collection without coordinated decision-making;
- do not involve an agricultural domain;
- provide insufficient methodological information.

Agent-based simulations may be included when agents explicitly make decisions, coordinate, negotiate, or interact over agricultural resources.

Review Workflow



7. Reference Management and PRISMA Workflow

Two complementary tools will be used rather than multiple overlapping reference managers.

Rayyan — SLR Screening and PRISMA Workflow

Rayyan will act as the primary systematic-review screening environment and authoritative record of:

- imported database records;
- duplicate identification and resolution;
- title/abstract screening;
- inclusion and exclusion decisions;
- full-text screening;
- exclusion reasons;
- screening status;
- PRISMA-related study counts.

Ambiguous duplicate matches will be manually inspected where necessary rather than automatically removed without verification.

The main deduplication process will therefore be carried out **once in Rayyan**, avoiding inconsistent duplicate counts across multiple applications.

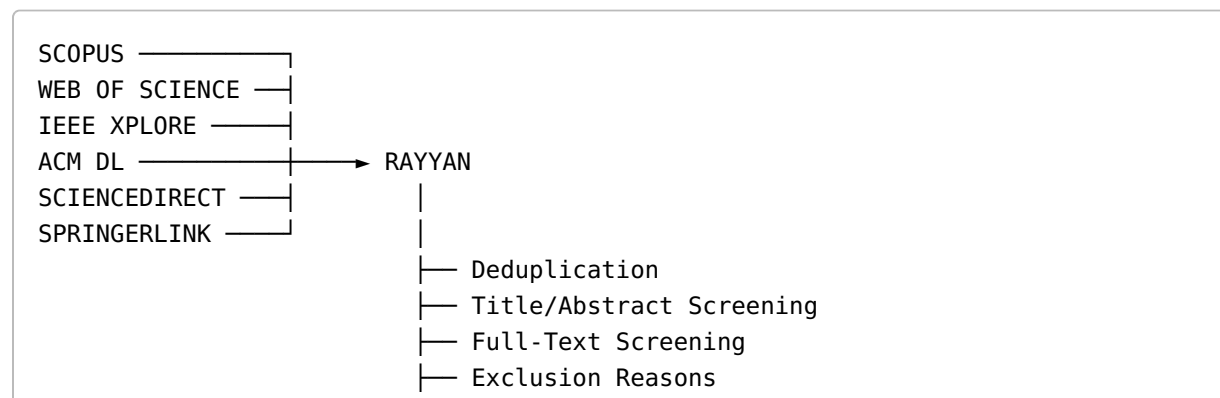
Zotero — Reference Library and Writing

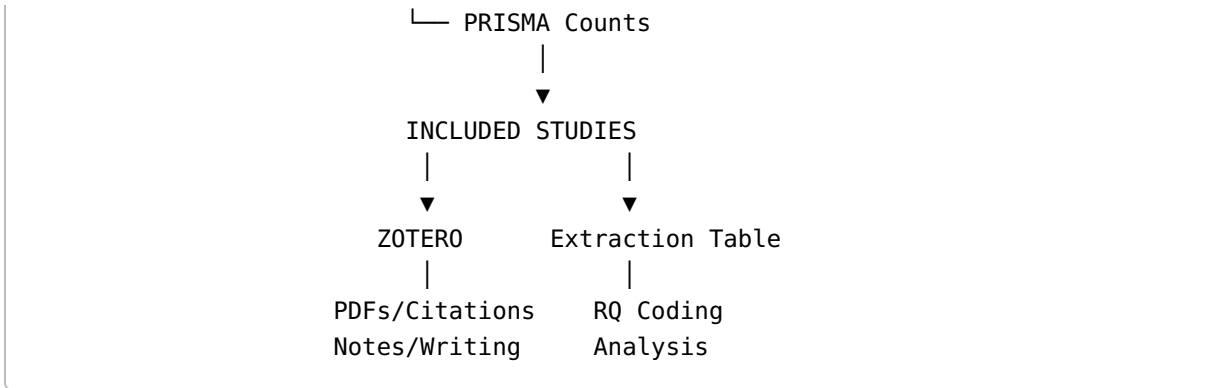
Zotero will be used for:

- permanent organisation of the literature;
- storage of included papers and PDFs;
- metadata management;
- annotations and research notes;
- citation insertion during report writing;
- bibliography generation;
- later reuse of the literature library for thesis work.

Mendeley and Zotero will **not both be used**, since they largely serve the same reference-management role and maintaining duplicate libraries would increase the risk of inconsistent metadata or version management.

The intended workflow is:





A separate structured extraction dataset, for example CSV or spreadsheet format, will contain one row per included study and the variables required to answer the research questions.

8. Quality Assessment and Data Extraction

Each included study will be assessed according to criteria such as:

1. clarity of research objectives;
2. clarity of agent roles and autonomy;
3. specification of communication and coordination mechanisms;
4. description of environment and data;
5. quality of experimental evaluation;
6. reproducibility;
7. acknowledgement of limitations.

A structured extraction table will record:

Dimension	Examples
Agricultural domain	irrigation, crops, livestock, mixed farming
Agent types	crop, water, weather, livestock, manager, machinery
Architecture	reactive, BDI, utility-based, hierarchical, distributed
Interaction	cooperation, negotiation, auction, contract-net
AI method	rules, optimisation, ML, RL/MARL, MPC, LLM
Resources	water, feed, energy, land, pasture, machinery
Conditions	drought, heatwave, rainfall, scarcity, uncertainty
Deployment	simulation, IoT, cyber-physical, field deployment
Agricultural outcomes	yield, crop stress, livestock health/welfare
Sustainability	water, GHG, energy, soil, pollution, land use

Dimension	Examples
Evaluation	datasets, metrics, baselines, scenarios
Limitations	scalability, uncertainty, interoperability, cost

9. Analysis and Synthesis

The literature will be compared across coordination strategies analogous to the original smart-farm concept:

1. **Independent/decentralised decision-making**
2. **Centralised or hierarchical coordination**
3. **Cooperative/negotiated multi-agent allocation**
4. **Learning or adaptive approaches**, where present

Reported outcomes will be mapped to common metric groups, including:

- water-use efficiency;
- unmet resource demand;
- crop yield, health, or stress;
- livestock welfare and heat stress;
- resource conflicts;
- resource-allocation fairness;
- farm-level utility or economic performance;
- communication and computational overhead;
- energy consumption;
- GHG or pollution reduction;
- soil, land, and environmental-resource efficiency.

The review will also examine scenarios corresponding to the proposed smart-farm system, including:

- normal agricultural conditions;
- water scarcity;
- severe drought;
- heatwaves;
- unexpected rainfall;
- simultaneous crop–livestock competition;
- environmental-resource constraints.

Where studies report sufficiently comparable quantitative outcomes, the findings will be summarised descriptively. Because significant heterogeneity in MAS architectures, agricultural systems, and evaluation metrics is expected, the main synthesis will use **structured thematic analysis, evidence mapping, and taxonomy construction** rather than assuming that a formal statistical meta-analysis will be possible.

10. Single-Reviewer Consideration

As the project will be conducted by one researcher, study selection will use a **single-reviewer screening process**. This will be acknowledged as a methodological limitation.

To improve consistency and transparency:

- inclusion and exclusion criteria will be defined before full screening;
- the criteria will be piloted on an initial sample of papers;
- exclusion reasons will be recorded systematically;
- ambiguous studies will be reconsidered during full-text review;
- a subset of records may be re-screened later to assess intra-reviewer consistency;
- all screening and PRISMA decisions will remain traceable in Rayyan.

11. Expected Outcomes and Future Thesis Foundation

The project is expected to produce:

1. a reproducible systematic review of MAS in smart agriculture;
2. a taxonomy of agricultural agents, architectures, and coordination mechanisms;
3. a detailed analysis of irrigation and shared-resource negotiation;
4. a synthesis of MAS and AI approaches to livestock management;
5. an environmental-sustainability evidence map covering water scarcity, GHG emissions, energy, pollution, soil, and land use;
6. comparison of independent, centralised, cooperative, and learning-based approaches;
7. identification of common datasets, scenarios, evaluation metrics, and deployment practices;
8. identification of technical and scientific research gaps;
9. a conceptual architecture for an **integrated Multi-Agent Smart Farm Decision System**;
10. evidence-based research hypotheses suitable for subsequent thesis investigation.

A possible future system, depending on the findings of the SLR, could involve crop, livestock, weather, and resource-management agents negotiating shared water and other resources under uncertain environmental conditions while explicitly considering:

agricultural productivity + animal welfare + water conservation + resource fairness + environmental impact

The SLR will therefore establish whether the integration of crop irrigation, livestock requirements, shared-resource negotiation, environmental sustainability, and adaptive multi-agent decision-making represents a defensible research gap rather than assuming that such a gap exists in advance.

12. Proposed Timeline

SEPTEMBER 2026

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- |— Finalise SLR protocol
- |— Finalise primary/secondary RQs
- |— Pilot database-specific search strings
- |— Execute database searches
- |— Export and merge records in Rayyan
- |— Deduplicate records
- |— Begin title/abstract screening

OCTOBER 2026

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- |— Complete title/abstract screening
- |— Full-text screening
- |— Record exclusion reasons
- |— Quality assessment
- |— Structured data extraction
- |— Taxonomy and evidence synthesis
- |— Environmental sustainability analysis
- |— Research-gap identification
- |— PRISMA documentation
- |— Conceptual future MAS architecture
- |— Final report and presentation

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PROJECT COMPLETION — OCTOBER 2026

Proposed Final Contribution

A systematic and reproducible synthesis of how Multi-Agent Systems are being used for agricultural coordination and decision-making, connecting crop irrigation, shared-resource negotiation, livestock management, and environmental sustainability, and using the resulting evidence to define the requirements and research gaps for a future integrated sustainable smart-farm MAS.